Домашнее задание 0

Обратная матрица

№ 1.4(37-42) методом присоединенной мат

№ 1.4.37

$$A=\begin{pmatrix}-1 & 0 & 0\\ 0 & 0 & 2\\ 0 & 0{,}5 & 0\end{pmatrix}$$

1) $\det A = \begin{pmatrix}-1 & 0 & 0\\ 0 & 0 & 2\\ 0 & 0{,}5 & 0\end{pmatrix} = (-1)\cdot 0\cdot 0 + 0\cdot 0{,}5\cdot 0 + 0\cdot 2\cdot 0 - 0\cdot 0\cdot 0 - 0{,}5\cdot 2\cdot(-1) - 0\cdot 0\cdot 2 = 1 \neq \Rightarrow \exists A^{-1}$

2) $A_{11} = (-1)^{1+1}\cdot\begin{vmatrix}0 & 2\\ 0{,}5 & 0\end{vmatrix} = 0\cdot 0 - 2\cdot 0{,}5 = -1$

$A_{12} = (-1)^{1+2}\cdot\begin{vmatrix}0 & 2\\ 0 & 0\end{vmatrix} = 0$

$A_{13} = (-1)^{1+3}\cdot\begin{vmatrix}0 & 0\\ 0 & 0{,}5\end{vmatrix} = 0$

$A_{21} = (-1)^{2+1}\cdot\begin{vmatrix}0 & 0\\ 0{,}5 & 0\end{vmatrix} = 0$

$A_{22} = (-1)^{2+2}\cdot\begin{vmatrix}-1 & 0\\ 0 & 0\end{vmatrix} = 0$

$A_{23} = (-1)^{2+3}\cdot\begin{vmatrix}-1 & 0\\ 0 & 0{,}5\end{vmatrix} = (-1)\cdot 0{,}5 - 0\cdot 0 = 0{,}5$

$A_{31} = (-1)^{3+1} \cdot \begin{vmatrix} 0 & 0 \\ 0 & 2 \end{vmatrix} = 0$

$A_{32} = (-1)^{3+2} \cdot \begin{vmatrix} -1 & 0 \\ 0 & 2 \end{vmatrix} = (-1 \cdot 2) - 0 \cdot 0 = 2$

$A_{33} = (-1)^{3+3} \cdot \begin{vmatrix} -1 & 0 \\ 0 & 0 \end{vmatrix} = (-1) \cdot 0 - 0 \cdot 0 = 0$

3) $\tilde{A} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0{,}5 \\ 0 & 2 & 0 \end{pmatrix}^T = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 0{,}5 & 0 \end{pmatrix}$

4 $A^{-1} = \frac{1}{\det(A)} \cdot \tilde{A} = \frac{1}{1} \cdot \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 0{,}5 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 0{,}5 & 0 \end{pmatrix}$

<u>1.4.38</u>

$A = \begin{pmatrix} 1 & 1 & -1 \\ 8 & 3 & -6 \\ -4 & -1 & 3 \end{pmatrix}$

1) $\det(A) = \begin{vmatrix} 1 & 1 & -1 \\ 8 & 3 & -6 \\ -4 & -1 & 3 \end{vmatrix} = 1 \cdot 3 \cdot 3 + 1 \cdot (-6) \cdot (-4) + 8 \cdot 1 \cdot 1 -$
$4 \cdot 3 \cdot 1 - 8 \cdot 1 \cdot 3 - 1 \cdot 6 \cdot 1 = -1 \neq 0 \Rightarrow \exists A^{-1}$

2) $A_{11} = (-1)^{1+1} \cdot \begin{vmatrix} 3 & -6 \\ -1 & 3 \end{vmatrix} = 3 \cdot 3 - 6 \cdot 1 = 3$

$$A_{12} = (-1)^{1+2} \cdot \begin{vmatrix} 8 & -6 \\ -4 & 3 \end{vmatrix} = -(8 \cdot 3 - (-6) \cdot (-4)) = 0$$

$$A_{13} = (-1)^{1+3} \cdot \begin{vmatrix} 8 & 3 \\ -4 & -1 \end{vmatrix} = 8 \cdot (-1) - (-4) \cdot 3 = 4$$

$$A_{21} = (-1)^{2+1} \cdot \begin{vmatrix} 1 & -1 \\ -1 & 3 \end{vmatrix} = -2$$

$$A_{22} = (-1)^{4} \cdot \begin{vmatrix} 1 & -1 \\ -4 & 3 \end{vmatrix} = -1$$

$$A_{23} = (-1)^{5} \cdot \begin{vmatrix} 1 & 1 \\ -4 & -1 \end{vmatrix} = -3$$

$$A_{31} = (-1)^{4} \cdot \begin{vmatrix} 1 & -1 \\ 3 & -6 \end{vmatrix} = 3$$

$$A_{32} = (-1)^{5} \cdot \begin{vmatrix} 1 & -1 \\ 8 & -6 \end{vmatrix} = -2$$

$$A_{33} = (-1)^{6} \cdot \begin{vmatrix} 1 & 1 \\ 8 & 3 \end{vmatrix} = -5$$

3) $\tilde{A} = \begin{pmatrix} 3 & 0 & 4 \\ -2 & -1 & -3 \\ 3 & -2 & -5 \end{pmatrix}^T = \begin{pmatrix} 3 & -2 & 3 \\ 0 & -1 & -2 \\ 4 & -3 & -5 \end{pmatrix}$

4) $A^{-1} = \frac{1}{\det(A)} \cdot \tilde{A} = \frac{1}{-1} \cdot \begin{pmatrix} 3 & -2 & 3 \\ 0 & -1 & -2 \\ 4 & -3 & -5 \end{pmatrix} = \begin{pmatrix} -3 & 2 & -3 \\ 0 & 1 & 2 \\ -4 & 3 & 5 \end{pmatrix}$

$$A_{32} = (-1)^5 \cdot \begin{vmatrix} 1 & 2 \\ 2 & 2 \end{vmatrix} = 2$$

$$A_{33} = (-\cancel{1})^6 \cdot \begin{vmatrix} 1 & 2 \\ 2 & 2 \end{vmatrix} = -3$$

3) $$\tilde{A} = \begin{pmatrix} -6 & 0 & 6 \\ -2 & -4 & 3 \\ 4 & 2 & -3 \end{pmatrix}^T = \begin{pmatrix} -6 & -2 & 4 \\ 0 & -4 & 2 \\ 6 & 3 & -3 \end{pmatrix}$$

4) $$A^{-1} = \frac{1}{\det(A)} \cdot \tilde{A} = \frac{1}{6} \cdot \begin{pmatrix} -6 & -2 & 4 \\ 0 & -4 & 2 \\ 6 & 3 & -3 \end{pmatrix} =$$

$$= \begin{pmatrix} \frac{-6}{6} & \frac{-2}{6} & \frac{4}{6} \\ 0 & \frac{-4}{6} & \frac{2}{6} \\ \frac{6}{6} & \frac{3}{6} & \frac{-3}{6} \end{pmatrix} = \begin{pmatrix} -1 & -\frac{1}{3} & \frac{2}{3} \\ 0 & -\frac{2}{3} & \frac{1}{3} \\ 1 & \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$$

11.4.40

$$A = \begin{pmatrix} 3 & 4 & 2 \\ 2 & -4 & -3 \\ 1 & 5 & 1 \end{pmatrix}$$

1) $$\det(A) = \begin{vmatrix} 3 & 4 & 2 \\ 2 & -4 & -3 \\ \dots \end{vmatrix} = 3 \cdot (-4) \cdot 1 + 4 \cdot (-3) \cdot 1 + 2 \cdot 5 \cdot 2 - 1 \cdot (-4) \cdot 2 - 3 \cdot 4 \cdot 1 - 5 \cdot (-3) \cdot 3 = 41 \neq 0 \Rightarrow$$

$\exists A^{-1}$

№ 1.4.39

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & -1 & 2 \\ 4 & 1 & 4 \end{pmatrix}$$

1) $\det(A) = \begin{vmatrix} 1 & 1 & 2 \\ 2 & -1 & 2 \\ 4 & 1 & 4 \end{vmatrix} = 1\cdot(-1)\cdot 4 + 1\cdot 2\cdot 4 + 2\cdot 1\cdot 2 - 4\cdot(-1)\cdot 2 - 2\cdot 1\cdot 4 - 1\cdot 1\cdot 2$

$\Rightarrow \exists A^{-1}$

2) $A_{11} = (-1)^2 \begin{vmatrix} -1 & 2 \\ 1 & 4 \end{vmatrix} = -6$

$A_{12} = (-1)^3 \begin{vmatrix} 2 & 2 \\ 4 & 4 \end{vmatrix} = 0$

$A_{13} = (-1)^4 \begin{vmatrix} 2 & -1 \\ 4 & 1 \end{vmatrix} = 6$

$A_{21} = (-1)^3 \begin{vmatrix} 1 & 2 \\ 1 & 4 \end{vmatrix} = -2$

$A_{22} = (-1)^4 \begin{vmatrix} 1 & 2 \\ 4 & 4 \end{vmatrix} = -4$

$A_{23} = (-1)^5 \begin{vmatrix} 1 & 1 \\ 4 & 1 \end{vmatrix} = 3$

$A_{31} = (-1)^4 \cdot \begin{vmatrix} 1 & 2 \\ -1 & 2 \end{vmatrix} = 4$

2) $A_{11} = \begin{vmatrix} -4 & -3 \\ 5 & 1 \end{vmatrix} = (-4)\cdot 1 + \ 11$

$A_{12} = -\begin{vmatrix} 2 & -3 \\ 1 & 1 \end{vmatrix} = -5$

$A_{13} = \begin{vmatrix} 2 & -4 \\ 1 & 5 \end{vmatrix} = 14$

$A_{21} = -\begin{vmatrix} 4 & 2 \\ 5 & 1 \end{vmatrix} = 6$

$A_{22} = \begin{vmatrix} 3 & 2 \\ 1 & 1 \end{vmatrix} = 1$

$A_{23} = -\begin{vmatrix} 3 & 4 \\ 1 & 5 \end{vmatrix} = -11$

$A_{31} = \begin{vmatrix} 4 & 2 \\ -4 & -3 \end{vmatrix} = -4$

$A_{32} = -\begin{vmatrix} 3 & 2 \\ 2 & -3 \end{vmatrix} = 13$

$A_{33} = \begin{vmatrix} 3 & 4 \\ 2 & -4 \end{vmatrix} = -20$

3) $\hat{A} = \begin{pmatrix} 11 & -5 & 14 \\ 6 & 1 & -11 \\ -4 & 13 & -20 \end{pmatrix}^T = \begin{pmatrix} 11 & 6 & -4 \\ -5 & 1 & 13 \\ 14 & -11 & -20 \end{pmatrix}$

4) $A^{-1} = \frac{1}{\det(A)} \cdot \tilde{A} = \frac{1}{41} \cdot \begin{pmatrix} 11 & 6 & -4 \\ -5 & 1 & 13 \\ 14 & -11 & -20 \end{pmatrix} =$

$$= \begin{pmatrix} \frac{11}{41} & \frac{6}{41} & \frac{-4}{41} \\ \frac{-5}{41} & \frac{1}{41} & \frac{13}{41} \\ \frac{14}{41} & -\frac{11}{41} & \frac{-20}{41} \end{pmatrix} = \begin{pmatrix} \frac{11}{41} & \frac{6}{41} & -\frac{4}{41} \\ -\frac{5}{41} & \frac{1}{41} & \frac{13}{41} \\ \frac{14}{41} & -\frac{11}{41} & -\frac{20}{41} \end{pmatrix}$$

№1.4.41

$$A = \begin{pmatrix} 3 & -1 & 2 \\ 4 & -3 & 3 \\ 1 & 3 & 0 \end{pmatrix}$$

1) $\det(A) = \begin{vmatrix} 3 & -1 & 2 \\ 4 & -3 & 3 \\ 1 & 3 & 0 \end{vmatrix} = 3 \cdot (-3) \cdot 0 + 4 \cdot 3 \cdot 2 + (-1) \cdot 3 \cdot 1 - 1 \cdot (-3) \cdot 2 - 0 - 3 \cdot 3 \cdot 3 = 0$

=> обратной матрицы не существует.

№1.4.42

$$A = \begin{pmatrix} 5 & 8 & -1 \\ 2 & -3 & 2 \\ 1 & 2 & 3 \end{pmatrix}$$

1.4.43

$$A = \begin{pmatrix} 1 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{pmatrix}$$

1) $\det(A) = 1\cdot 2\cdot 2 + (-1)\cdot 1\cdot(-1) + (-1)\cdot(-1)\cdot 1 - (-1)\cdot 2(-1)$
$\quad - 1\cdot 1\cdot (1) - (-1)(-1)\cdot 2 = 1 \neq 0 \Rightarrow$

2) $$(A|E) = \left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ -1 & 2 & 1 & 0 & 1 & 0 \\ -1 & 1 & 2 & 0 & 0 & 1 \end{array}\right) \begin{array}{l} \\ II + I \\ III + I \end{array} \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array}\right) \begin{array}{l} I + III \\ \\ \end{array} \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & -1 & 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array}\right) \begin{array}{l} I + II \\ \\ \end{array} \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array}\right) = (E|A^{-1})$$

$$A^{-1} = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

1) $\det(A) = \begin{vmatrix} 5 & 8 & -1 \\ 2 & -3 & 2 \\ 1 & 2 & 3 \end{vmatrix} = 5\cdot(-3)\cdot 3 + 8\cdot 2\cdot 1 + 2\cdot 2\cdot(-1)$
$-(-1)\cdot(-3)\cdot(-1) - 8\cdot 2\cdot 3 -$
$= -104 \neq 0 \Rightarrow \exists A^{-1}$

2) $A_{11} = \begin{vmatrix} -3 & 2 \\ 2 & 3 \end{vmatrix} = -13$

$A_{12} = -\begin{vmatrix} 2 & 2 \\ 1 & 3 \end{vmatrix} = -4$

$A_{13} = \begin{vmatrix} 2 & -3 \\ 1 & 2 \end{vmatrix} = 7$

$A_{21} = -\begin{vmatrix} 8 & -1 \\ 2 & 3 \end{vmatrix} = -26$

$A_{22} = \begin{vmatrix} 5 & -1 \\ 1 & 3 \end{vmatrix} = +16$

$A_{23} = -\begin{vmatrix} 5 & 8 \\ 1 & 2 \end{vmatrix} = -2$

$A_{31} = \begin{vmatrix} 8 & -1 \\ -3 & 2 \end{vmatrix} = 13$

$A_{32} = -\begin{vmatrix} 5 & -1 \\ 2 & 2 \end{vmatrix} = -12$

$A_{33} = \begin{vmatrix} 5 & 8 \\ 2 & -3 \end{vmatrix} = -31$

1.4 [illegible]

$$A=\begin{pmatrix}1 & 2 & -2 & 4\\ 2 & 6 & 1 & 0\\ 3 & 0 & 1 & 2\\ -1 & 4 & 5 & -4\end{pmatrix}$$

$$\det(A) = -2\begin{vmatrix}2 & -2 & 4\\ 0 & 1 & 2\\ 4 & 5 & -4\end{vmatrix} + 6\begin{vmatrix}1 & -2 & 4\\ 3 & 1 & 2\\ -1 & 5 & -4\end{vmatrix} - 1\begin{vmatrix}1 & 2 & 4\\ 3 & 0 & 2\\ -1 & 4 & -4\end{vmatrix} + 0 = -2(-60)+6\cdot 30 - 1\cdot 60 = 240 \neq 0 \Rightarrow \exists A^{-1}$$

$$(A|E)=\left(\begin{array}{cccc|cccc}1 & 2 & -2 & 4 & 1 & 0 & 0 & 0\\ 2 & 6 & 1 & 0 & 0 & 1 & 0 & 0\\ 3 & 0 & 1 & 2 & 0 & 0 & 1 & 0\\ -1 & 4 & 5 & -4 & 0 & 0 & 0 & 1\end{array}\right)\begin{array}{l}\\ II-2I\\ III-3\cdot I\\ IV+I\end{array} \sim$$

$$\sim\left(\begin{array}{cccc|cccc}1 & 2 & -2 & 4 & 1 & 0 & 0 & 0\\ 0 & 2 & 5 & -8 & -2 & 1 & 0 & 0\\ 0 & -6 & 7 & -10 & -3 & 0 & 1 & 0\\ 0 & 6 & 3 & 0 & 1 & 0 & 0 & 1\end{array}\right)\begin{array}{l}\\ \\ III+3II\\ IV-3II\end{array} \sim$$

$$\sim\left(\begin{array}{cccc|cccc}1 & 2 & -2 & 4 & 1 & 0 & 0 & 0\\ 0 & 2 & 5 & -8 & -2 & 1 & 0 & 0\\ 0 & 0 & 22 & -34 & -9 & 3 & 1 & 0\\ 0 & 0 & -12 & 24 & 7 & -3 & 0 & 1\end{array}\right)\begin{array}{l}[illegible]\\ II:2\end{array} \sim$$

$2\cdot(-3)-$

$2\cdot 3 - 2\cdot 2\cdot 3$

3) $\tilde{A} = \begin{pmatrix} -13 & -4 & -7 \\ -26 & 16 & -2 \\ 13 & -12 & -31 \end{pmatrix}^T = \begin{pmatrix} -13 & -26 & 13 \\ -4 & 16 & -12 \\ -7 & -2 & -31 \end{pmatrix}$

4) $A^{-1} = \frac{1}{\det(A)} \cdot \tilde{A} = -\frac{1}{104} \cdot \begin{pmatrix} +\frac{13}{104} & +\frac{26}{104} & -\frac{13}{104} \\ \frac{4}{104} & -\frac{16}{104} & \frac{12}{104} \\ \frac{7}{104} & \frac{2}{104} & \frac{31}{104} \end{pmatrix} =$

$= \begin{pmatrix} \frac{1}{8} & \frac{1}{4} & -\frac{1}{8} \\ \frac{1}{26} & -\frac{2}{13} & \frac{3}{26} \\ \frac{7}{104} & \frac{1}{52} & \frac{31}{104} \end{pmatrix}$

1.4(43-45) найти методом элементарных преобразований)

Пр.

$$\sim\left(\begin{array}{cccc|cccc} 1 & 2 & -2 & 4 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2{,}5 & -4 & -1 & 0{,}5 & 0 & 0 \\ 0 & 0 & -22 & -34 & -9 & 3 & 1 & 0 \\ 0 & 0 & -12 & 24 & 4 & -3 & 0 & 1 \end{array}\right)\begin{array}{l} I-2II \\ \\ \\ 11\cdot IV+6III \end{array}\sim$$

$$\sim\left(\begin{array}{cccc|cccc} 1 & 0 & -7 & 12 & 3 & -1 & 0 & 0 \\ 0 & 1 & 2{,}5 & -4 & -1 & 0{,}5 & 0 & 0 \\ 0 & 0 & 22 & -34 & -9 & 3 & 1 & 0 \\ 0 & 0 & 0 & 60 & 23 & -15 & 6 & 11 \end{array}\right)\begin{array}{l} \\ \\ III:22 \\ IV:60 \end{array}\sim$$

$$\sim\left(\begin{array}{cccc|cccc} 1 & 0 & -7 & 12 & 3 & -1 & 0 & 0 \\ 0 & 1 & 2{,}5 & -4 & -1 & 0{,}5 & 0 & 0 \\ 0 & 0 & 1 & -\frac{17}{11} & -\frac{9}{22} & \frac{3}{22} & \frac{1}{22} & 0 \\ 0 & 0 & 0 & 1 & \frac{23}{60} & -\frac{15}{60} & \frac{1}{10} & \frac{11}{60} \end{array}\right)\begin{array}{l} I+7\cdot III-\frac{13}{11}IV \\ II+4\cdot IV \\ III+\frac{17}{11}\cdot IV \\ \\ \end{array}\sim$$

$$\sim\left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -\frac{19}{60} & \frac{1}{4} & \frac{1}{5} & -\frac{13}{60} \\ 0 & 1 & \frac{5}{2} & 0 & \frac{8}{15} & -\frac{1}{2} & \frac{2}{5} & \frac{11}{15} \\ 0 & 0 & 1 & 0 & \frac{11}{60} & -\frac{1}{4} & \frac{1}{5} & \frac{17}{60} \\ 0 & 0 & 0 & 1 & \frac{23}{60} & -\frac{15}{60} & \frac{1}{10} & \frac{11}{60} \end{array}\right)\begin{array}{l} \\ II-\frac{5}{2}III \\ \\ \\ \end{array}\sim$$

$$\sim\left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -\frac{19}{60} & \frac{1}{4} & \frac{1}{5} & -\frac{13}{60} \\ 0 & 1 & 0 & 0 & \frac{3}{40} & \frac{1}{8} & -\frac{1}{10} & \frac{1}{40} \\ 0 & 0 & 1 & 0 & \frac{11}{60} & -\frac{1}{4} & \frac{1}{5} & \frac{17}{60} \\ 0 & 0 & 0 & 1 & \frac{23}{60} & -\frac{15}{60} & \frac{1}{10} & \frac{11}{60} \end{array}\right) =$$

3) $X = B \cdot A^{-1} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix} =$

$= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

№ 1.4.52

$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \cdot x = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

$x = A^{-1} \cdot B$

1) $\det(A) = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 0 \Rightarrow$ обратной матрицы не существует

Ответ: x не существует

№ 1.4.53

$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} x = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$

$\det A = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 0 \Rightarrow$ не сущ

Ответ: x не существует.

$(E \mid A^{-1})$.

$$A^{-1} = \begin{pmatrix} -\frac{19}{60} & \frac{1}{4} & \frac{1}{6} & -\frac{13}{60} \\ \frac{3}{40} & \frac{1}{8} & -\frac{1}{10} & \frac{1}{40} \\ \frac{11}{60} & -\frac{1}{4} & \frac{1}{6} & \frac{17}{60} \\ \frac{\cancel{24}\,23}{60} & -\frac{19}{60} & \frac{1}{10} & \frac{11}{60} \end{pmatrix}$$

1.4 (40 – 58) решить

1.4.40

$$X \cdot \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

т.к. $XA = B \Rightarrow X = B \cdot A^{-1}$

$$\det(A) = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = -2 \neq 0 \Rightarrow \exists A^{-1}$$

$$A^{-1} = \frac{1}{\det A} \cdot \tilde{A} = \frac{1}{-2} \cdot \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$$

1.4.56

$$X \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{pmatrix}$$

$$X = B \cdot A^{-1}$$

1) $\det(A) = \cancel{1\ 0\ 0}\ 6 \neq 0 \Rightarrow \exists A^{-1}$

2) $$(A|E) = \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 3 & 0 & 0 & 1 \end{array}\right) \begin{array}{l} \\ II:2 \\ III:3 \end{array} \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0{,}5 & 0 \\ 0 & 0 & 1 & 0 & 0 & \frac{1}{3} \end{array}\right) = (E|A)$$

$$A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{pmatrix}$$

3) $\cancel{A} = X = B \cdot A^{-1} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{pmatrix} =$

$$=\begin{pmatrix} 1\cdot 0 & 0 & 1\cdot\frac{1}{3} \\ 0 & 0.5\cdot 2 & 0 \\ 3\cdot 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & \frac{1}{3} \\ 0 & 1 & 0 \\ 3 & 0 & 0 \end{pmatrix}$$

<u>14.58</u>

$$\begin{pmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ 0 & -2 & 1 \end{pmatrix} \cdot X \cdot \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{pmatrix}$$

$$x = A^{-1} \cdot B \cdot C^{-1}$$

1) $\det(A) = 3 + 0 - 12 - 0 + 4 - 2 = -7 \neq 0 \Rightarrow \exists A^{-1}$

$\det(C) = 0 + 84 + 96 - 105 - 48 = 27 \neq 0 \Rightarrow \exists A^{-1}$

2) $(A|E) = \left(\begin{array}{ccc|ccc} 1 & -2 & 3 & 1 & 0 & 0 \\ 2 & 3 & -1 & 0 & 1 & 0 \\ 0 & -2 & 1 & 0 & 0 & 1 \end{array}\right)$ $\begin{matrix} I \\ II - 2I \end{matrix}$ $\sim$

$\sim \left(\begin{array}{ccc|ccc} 1 & -2 & 3 & 1 & 0 & 0 \\ 0 & 7 & -5 & -2 & 1 & 0 \\ 0 & -2 & 1 & 0 & 0 & 1 \end{array}\right)$ I·

$)=$

№1.54

$$\begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} \cdot X \cdot \begin{pmatrix} -5 & 6 \\ -4 & 5 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}$$

$AXC = B \Rightarrow X = A^{-1} \cdot B \cdot C^{-1}$

1) $\det(A) = 5 \neq 0 \Rightarrow \exists A^{-1}$

$\det(C) = \begin{vmatrix} -5 & 6 \\ -4 & 5 \end{vmatrix} = -1 \neq 0 \Rightarrow \exists A^{-1}$

2) $A^{-1} = \frac{\tilde{A}}{\det(A)} = \frac{1}{5} \cdot \begin{pmatrix} 3 & 1 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} \frac{3}{5} & \frac{1}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{pmatrix}$

$C^{-1} = \frac{\tilde{C}}{\det(C)} = -1 \begin{pmatrix} 5 & -6 \\ 4 & -5 \end{pmatrix} = \begin{pmatrix} -5 & 6 \\ -4 & 5 \end{pmatrix}$

3) $X = \begin{pmatrix} \frac{3}{5} & \frac{1}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{pmatrix} \cdot \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} \cdot \begin{pmatrix} -5 & 6 \\ -4 & 5 \end{pmatrix} =$

$= \begin{pmatrix} \frac{3}{5} + \frac{1}{5} \cdot 2 & -\frac{3}{5} + \frac{3}{5} \\ -\frac{2}{5} + \frac{2}{5} & \frac{2}{5} + \frac{3}{5} \end{pmatrix} \cdot \begin{pmatrix} -5 & 6 \\ -4 & 5 \end{pmatrix} =$

$= \begin{pmatrix} 1 \cdot (-5) + 0 & 6 \\ 0 + 1 \cdot (-4) & 5 \end{pmatrix} = \begin{pmatrix} -5 & 6 \\ -4 & 5 \end{pmatrix}$

решы не

$$= \begin{pmatrix} \frac{1}{7} & \frac{4}{7} & \frac{1}{7} \\ \frac{2}{7} & \frac{1}{7} & -\frac{7}{7} \\ \frac{4}{7} & -\frac{2}{7} & -\frac{7}{7} \end{pmatrix} = \begin{pmatrix} -\frac{1}{7} & \frac{4}{7} & 1 \\ \frac{2}{7} & -\frac{1}{7} & -1 \\ \frac{4}{7} & -\frac{2}{7} & -1 \end{pmatrix}$$

3) $C_{11} = \begin{vmatrix} 5 & 6 \\ 8 & 0 \end{vmatrix} = -48$

$C_{12} = -\begin{vmatrix} 4 & 6 \\ 7 & 0 \end{vmatrix} = -3$

$C_{13} = \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} = -3$

$C_{21} = -\begin{vmatrix} 2 & 3 \\ 8 & 0 \end{vmatrix} = 24$

$C_{22} = \begin{vmatrix} 1 & 3 \\ 7 & 0 \end{vmatrix} = -21$

$C_{23} = -\begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix} = 6$

$C_{31} = \begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} = -3$

$C_{32} = -\begin{vmatrix} 1 & 3 \\ 4 & 6 \end{vmatrix} = 6$

2) $A_{11} = \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix} = 1$

$A_{12} = -\begin{vmatrix} 2 & -1 \\ 0 & 1 \end{vmatrix} = -2$

$A_{13} = \begin{vmatrix} 2 & 3 \\ 0 & -2 \end{vmatrix} = -4$

$A_{21} = -\begin{vmatrix} -2 & 3 \\ -2 & 1 \end{vmatrix} = -4$

$A_{22} = \begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix} = 1$

$A_{23} = -\begin{vmatrix} 1 & -2 \\ 0 & -2 \end{vmatrix} = 2$

$A_{31} = \begin{vmatrix} -2 & 3 \\ 3 & -1 \end{vmatrix} = -7$

$A_{32} = -\begin{vmatrix} 1 & 3 \\ 2 & -1 \end{vmatrix} = 7$

$A_{33} = \begin{vmatrix} 1 & -2 \\ 2 & 3 \end{vmatrix} = 7$

$$\tilde{A} = \begin{pmatrix} 1 & -2 & -4 \\ -4 & 1 & 2 \\ -7 & 7 & 7 \end{pmatrix}^{T} = \begin{pmatrix} 1 & -4 & -7 \\ -2 & 1 & 7 \\ -4 & 2 & 7 \end{pmatrix}$$

$$A^{-1} = \frac{\tilde{A}}{\det A} =$$

$8x_1 + 6x_2 + 5x_3$

$$\left(\begin{array}{cccc|c} 8 & 6 & 5 & 2 & 21 \\ 3 & 3 & 2 & 1 & 10 \\ 4 & 2 & 3 & 1 & 8 \\ 3 & 5 & 1 & 1 & 15 \\ 7 & 4 & 5 & 2 & 18 \end{array}\right)_{IV - II} \sim$$

$$\sim \left(\begin{array}{cccc|c} 8 & 6 & 5 & 2 & 21 \\ 3 & 3 & 2 & 1 & 10 \\ 4 & 2 & 3 & 1 & 8 \\ 0 & 2 & -1 & 0 & 5 \\ 7 & 4 & 5 & 2 & 18 \end{array}\right)$$

мяу

$$\tilde{C} = \begin{pmatrix} -48 & 42 & -3 \\ 24 & -21 & 6 \\ -3 & 6 & -3 \end{pmatrix}^T = \begin{pmatrix} -48 & 24 & -3 \\ 42 & -21 & 6 \\ -3 & 6 & -3 \end{pmatrix}$$

$$C^{-1} = \frac{\tilde{C}}{\det(C)} = \begin{pmatrix} -\frac{48}{27} & \frac{24}{27} & -\frac{3}{27} \\ \frac{42}{27} & -\frac{21}{27} & \frac{6}{27} \\ -\frac{3}{27} & \frac{6}{27} & -\frac{3}{27} \end{pmatrix} =$$

$$= \begin{pmatrix} -\frac{16}{9} & \frac{8}{9} & -\frac{1}{9} \\ \frac{14}{9} & -\frac{7}{9} & \frac{2}{9} \\ -\frac{1}{9} & \frac{2}{9} & -\frac{1}{9} \end{pmatrix}$$

4) $X = A^{-1} \cdot B \cdot C^{-1} =$

$$= \begin{pmatrix} -\frac{1}{7} & \frac{4}{7} & 1 \\ \frac{2}{7} & -\frac{1}{7} & -1 \\ \frac{4}{7} & -\frac{2}{7} & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{pmatrix} \cdot \begin{pmatrix} -\frac{16}{9} & \frac{8}{9} & -\frac{1}{9} \\ \frac{14}{9} & -\frac{7}{9} & \frac{2}{9} \\ -\frac{1}{9} & \frac{2}{9} & -\frac{1}{9} \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{1}{7} & \frac{4}{7} & 1 \\ \frac{2}{7} & -\frac{1}{7} & -1 \\ \frac{4}{7} & -\frac{2}{7} & -1 \end{pmatrix}$$